

INTERNATIONAL GCSE

Biology

9201/2

Paper 2

Mark Scheme

November 2023

Version: 1.0 Final



Mark schemes are prepared by the Lead Assessment Writer and considered, together with the relevant questions, by a panel of subject teachers. This mark scheme includes any amendments made at the standardisation events which all associates participate in and is the scheme which was used by them in this examination. The standardisation process ensures that the mark scheme covers the students' responses to questions and that every associate understands and applies it in the same correct way. As preparation for standardisation each associate analyses a number of students' scripts. Alternative answers not already covered by the mark scheme are discussed and legislated for. If, after the standardisation process, associates encounter unusual answers which have not been raised they are required to refer these to the Lead Examiner.

It must be stressed that a mark scheme is a working document, in many cases further developed and expanded on the basis of students' reactions to a particular paper. Assumptions about future mark schemes on the basis of one year's document should be avoided; whilst the guiding principles of assessment remain constant, details will change, depending on the content of a particular examination paper.

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Level of response marking instructions

Level of response mark schemes are broken down into levels, each of which has a descriptor. The descriptor for the level shows the average performance for the level. There are marks in each level.

Before you apply the mark scheme to a student's answer read through the answer and annotate it (as instructed) to show the qualities that are being looked for. You can then apply the mark scheme.

Step 1 Determine a level

Start at the lowest level of the mark scheme and use it as a ladder to see whether the answer meets the descriptor for that level. The descriptor for the level indicates the different qualities that might be seen in the student's answer for that level. If it meets the lowest level then go to the next one and decide if it meets this level, and so on, until you have a match between the level descriptor and the answer. With practice and familiarity you will find that for better answers you will be able to quickly skip through the lower levels of the mark scheme.

When assigning a level you should look at the overall quality of the answer and not look to pick holes in small and specific parts of the answer where the student has not performed quite as well as the rest. If the answer covers different aspects of different levels of the mark scheme you should use a best fit approach for defining the level and then use the variability of the response to help decide the mark within the level, ie if the response is predominantly level 3 with a small amount of level 4 material it would be placed in level 3 but be awarded a mark near the top of the level because of the level 4 content.

Step 2 Determine a mark

Once you have assigned a level you need to decide on the mark. The descriptors on how to allocate marks can help with this. The exemplar materials used during standardisation will help. There will be an answer in the standardising materials which will correspond with each level of the mark scheme. This answer will have been awarded a mark by the Lead Examiner. You can compare the student's answer with the example to determine if it is the same standard, better or worse than the example. You can then use this to allocate a mark for the answer based on the Lead Examiner's mark on the example.

You may well need to read back through the answer as you apply the mark scheme to clarify points and assure yourself that the level and the mark are appropriate.

Indicative content in the mark scheme is provided as a guide for examiners. It is not intended to be exhaustive and you must credit other valid points. Students do not have to cover all of the points mentioned in the Indicative content to reach the highest level of the mark scheme.

An answer which contains nothing of relevance to the question must be awarded no marks.

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.1	cytoplasm		1	AO1 3.1.1a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.2	they have a nucleus		1	AO1 3.1.1c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
1.3	(A) cell wall		1	AO1 3.1.1c
	(B) cell membrane	allow plasma membrane	1	
	(C) plasmid	allow (small) loop of DNA	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.4	any two from: - it has a (large / permanent) vacuole - it has a cell wall - it has chloroplasts		2	AO2 3.1.1a 3.1.1b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
01.5	131.9 ÷ 4.6		1	AO2 3.1.1b 3.1.1c 6.3.3 6.3.4
	28.6739....		1	
	28.7		1	

Total			10	
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.1	$6400 \div 16\,000 \times (100)$	allow $0.4 \times (100)$	1	AO2 3.4.7 6.3.3
	40 % ()		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.2	y-axis labelled 'percentage of bacteria that survived'	1 or 2 bars correct = 1 mark	1	AO2 3.4.7 6.3.9 6.3.11
	3 bars correct		2	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.3	N		1	AO3 3.4.7

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.4	any two from: - sterilise tubes before use - sterilise liquid culture medium - sterilise measuring device (to transfer liquids) - flame the mouths of tubes when they are opened - stopper each tube with sterile cotton wool balls	allow description of how to do techniques } if these marks not awarded allow 1 mark for sterilise equipment	2	AO4 3.4.7 6.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.5	10 cm ³ pipette		1	AO4 3.4.7 6.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
02.6	2 mm = 2000 micrometres	allow conversion at any stage of calculation	1	AO2 3.4.7 6.3.3
	2000 ÷ 10		1	
	(× 200)		1	

Total			12
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.1	Tissue Name of the tissue	all correct = 2 marks 1 or 2 correct = 1 mark allow lines from Figure 4 to correct names	2	AO2 3.1.3a, 3.1.3b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.2	(epithelial tissue) covers / protects the (internal / external) surfaces (muscular tissue) <u>contracts</u> to move the food (around / through the stomach)		1 1	AO1 3.1.3a 3.1.3b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.3	1		1	AO2 3.2.4b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.4	3		1	AO2 3.2.4b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.5	(because the shape of the) <u>active site</u> has been changed or (because) the <u>active site</u> will not fit any food molecules		1	AO2 3.2.4b

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.6	absorbs water (and produces faeces)	ignore nutrients	1	AO1 3.1.3c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.7	produces bile	allow it makes the small intestine alkaline for lipase action	1	AO1 3.1.3c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
03.8	will not digest fats / lipids properly	allow small intestine will not be alkaline enough for efficient digestion	1	AO3 3.2.4c

Total		10
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.1	$6\text{H}_2\text{O} + 6\text{CO}_2 \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$		1	AO1 3.2.1a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.2	any two from: • for respiration • to produce starch • to produce fat / oil / lipid • to produce cellulose • to produce protein	do not accept energy produced / made / created allow to produce fatty acids or to produce glycerol allow to produce amino acid or to produce named protein	2	AO1 3.2.1 d

Question	Answers	Mark	AO/ Spec. Ref.
04.3	Level 3: The method would lead to the production of a valid outcome. All key steps are identified and logically sequenced	5–6	AO4 6.2.2 3.2.1c
	Level 2: The method would not necessarily lead to a valid outcome. Most steps are identified, but the plan is not fully logically sequenced.	3–4	
	Level 1: The method would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.	1–2	
	No relevant content	0	
	Indicative content • change the temperature • five or more temperatures • control the temperature using a water bath or kettle • use a thermometer to check the temperature • count the number of bubbles or measure the volume of oxygen / gas produced • (measure oxygen / gas produced) in a set period of time • use a stopwatch / timer • control colour of light • control light intensity • by keeping the light at the same distance • use the same (type / species of) pondweed • use the same length / size of pondweed • control carbon dioxide supply • idea of allowing time for pondweed to equilibrate • repeat each test two or more times • calculate a mean For Level 3 the method described must include, how the temperature is changed, the measurements needed to determine the rate of photosynthesis and control variables.		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.4	temperature		1	AO4 6.2.2 3.2.1c

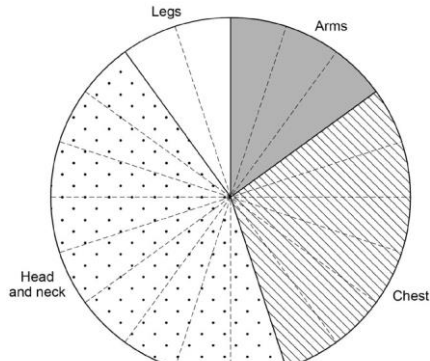
Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.5	62 (100 lux, test 2)		1	AO4 6.2.2 3.2.1c

Question	Answers	Extra information	Mark	AO / Spec. Ref.
04.6	any one from: • discard the result • do not use it for the mean	allow repeat the test	1	AO4 6.2.2 3.2.1c

Total		12
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.1	thermoregulatory centre	allow hypothalamus	1	AO1 3.4.4a

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.2	100 – 15 + 30 + (45)		1	AO2 3.4.4c
	10 % ()		1	6.3.3

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.3		all sections correct size labels correct	1 1	AO2 3.4.4c 6.3.9

Question	Answers	Extra information	Mark	AO / Spec. Ref.
05.4	any three from: • all skin temperatures increase • the skin temperature is always highest in the head • the increase in head temperature is less than the hands and feet • temperature (of all parts) rises most rapidly / steeply at the start • the change in rate of feet temperature increase occurs at 28 °C, but in the hands and head it is a more gradual change	allow skin temperature is always lowest in the feet allow increase in feet temperature is greater than the hands and head. allow temperature (of all parts) rises most rapidly / steeply between 23 °C–25/26 °C allow rate of temperature rise for all parts eventually slows down	3	AO3 3.4.4

Total		8
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.1	ethanol	in either order ignore alcohol	1	AO1 3.2.6m
	carbon dioxide		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
06.2	no / reduced oxygen / air	ignore references to temperature	1	AO1 3.2.6m

Question	Answers	Mark	AO/ Spec. Ref.
06.3	Level 3: A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.	5–6	AO3 3.2.6m
	Level 2: Some logically linked reasons are given. There may also be a simple judgement.	3–4	
	Level 1: Relevant points are made. They are not logically linked.	1–2	
	No relevant content	0	
	Indicative content: Pro: • A shows the largest decrease in pH • the pH has decreased as carbon dioxide is produced (in anaerobic respiration) • so statement is supported • the SD for A does not overlap with either B or C • so the difference is significant • 20 is a large number of repeats • so the results are representative Con: • the SDs for A (and C) overlap with D • so the difference is not significant • carbon dioxide is also produced in aerobic respiration • which would decrease the pH (of the solution) • no information has been given about control variables • so results may not be valid • only the SD for B does not overlap with any other species • therefore B has significantly less pH change than the other species For Level 3 an inclusion of a discussion of significance is required.		
Total		9	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.1	water glucose haemoglobin DNA	water → glucose glucose → haemoglobin haemoglobin → DNA	1 1 1	AO2 3.5.3h 6.3.1

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.2	double helix		1	AO2 3.5.3j

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.3	(3 bases code for one amino acid in the protein chain) the sequence of bases in the DNA determines the sequence of amino acids the sequence of amino acids determines which particular / different / specific protein is formed		1 1	AO1 3.5.3k

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.4	any one from: • painful joints • arthritis • unable to move freely • joints wear away	allow painful bone (in the joint)	1	AO3 3.5.3k

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.5	(Person) Z	mark only awarded if person Z is identified	1	AO3 3.5.3k
	(Reason) they have the stop code ACT (as the fourth amino acid code) so short / faulty C2 will be formed		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.6	(error in the egg cell) if the egg had two X chromosomes and the sperm had one Y it would result in XXY		1	AO3 3.5.4b
	(error in the sperm cell) if the egg had one X and the sperm had an X and a Y this would result in XXY		1	
		if no other marks awarded allow one mark for there was an error in meiosis when the gamete / egg / sperm was formed		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.7	copies of the DNA / genetic material are made including copies of the XXY chromosomes		1	AO2 3.5.2e, 3.5.2i
	(therefore) each new body cell receives one set / copy of the XXY chromosomes when the cell divides		1	
		allow the cell divides (once) to form two cells identical to each other with XXY chromosomes		

Question	Answers	Extra information	Mark	AO / Spec. Ref.
07.8		allow converse if clearly referring to sperm cell		AO2 3.5.3d
	the chromosomes are in pairs	allow there are 46 chromosomes	1	
	sperm cells are only in men / males and XX shows it is a woman / female	allow there is no Y chromosome to show it is a man / male	1	

Total		15
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Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.1	negative feedback (control)	ignore homeostasis	1	AO2 3.4.3g

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.2	pituitary gland monitors / detects water content of the blood		1	AO1 3.4.3e
	(and pituitary gland) secretes ADH if the water content is low		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.3	in the blood (stream)	allow in the plasma	1	AO1 3.4.3e

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.4	suitable scale and correct label on x axis	linear scale occupying at least half of grid	1	AO2 3.4.3f 6.3.9 6.3.11
	all points correct	allow tolerance of $\pm$ half a small square allow 1 mark for 4 or 5 correct points	2	
	smooth curve of best fit		1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.5	almost maximum reduction in blood pressure	allow 100 mg/day gives small extra reduction	1	AO3 3.4.3f
	100 mg/day is more likely to cause side effects	allow any side effects would be worse at 100 mg/day	1	

Question	Answers	Extra information	Mark	AO / Spec. Ref.
08.6	more sodium ions will be in the liquid / urine / filtrate (in the kidney tubules)	allow more sodium ions will be lost from the blood / body	1	AO3 3.4.3f 3.1.5e
	which increases the concentration (of the liquid / urine)	allow it decreases the water potential (of the liquid / urine)	1	
	which causes water to be kept in the liquid / urine instead of being reabsorbed	allow which leads to less water being reabsorbed	1	
	(so) blood volume decreases (and blood pressure decreases)		1	
Total			14	